

Branch: gh-pages genstats / inst / doc / 02\_03\_dimension-reduction.Rmd

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1 contributor

167 lines (116 sloc) 4.3 KB

RawBlameHistory

| title                            | author    | output                   |                                                                                                                      |
|----------------------------------|-----------|--------------------------|----------------------------------------------------------------------------------------------------------------------|
| Dimension reduction for genomics | Jeff Leek | rmarkdown::html_document | vignette                                                                                                             |
|                                  |           | toc                      |                                                                                                                      |
|                                  |           | true                     | %\VignetteIndexEntry{Dimension reduction for genomics}%\VignetteEngine{knitr::rmarkdown} \usepackage{utf8}{inputenc} |

# Dependencies

This document depends on the following packages:

```
suppressPackageStartupMessages({
  library(devtools)
  library(Biobase)
})
```

```
library(devtools)
library(Biobase)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase"))
```

# General principles

- Can we find patterns in matrices of data?

# Load some data

We will use this expression set that combines two studies Transcriptome genetics using second generation sequencing in a Caucasian population. and Understanding mechanisms underlying human gene expression variation with RNA sequencing.. These studies are different populations but we counted the same genes for both. Then we'll explore the differences.

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/montpick_eset.RData")
load(file=con)
close(con)
mp = montpick.eset
```

```
pdata=pData(mp)
edata=as.data.frame(exprs(mp))
fdata = fData(mp)
ls()
```

## Calculate the singular vectors

Here we calculate the singular vectors:

```
edata = edata[rowMeans(edata) > 100, ]
edata = log2(edata + 1)
edata_centered = edata - rowMeans(edata)
svd1 = svd(edata_centered)
names(svd1)
```

## Look at the percent variance explained

The percent of variance explained is given by  $\frac{d_{ii}}{\sum_j d_{jj}^2}$

```
plot(svd1$d,ylab="Singular value",col=2)
plot(svd1$d^2/sum(svd1$d^2),ylab="Percent Variance Explained",col=2)
```

## Plot top two principal components

```
par(mfrow=c(1,2))
plot(svd1$v[,1],col=2,ylab="1st PC")
plot(svd1$v[,2],col=2,ylab="2nd PC")
```

## Plot PC1 vs. PC2

A very common plot is to plot PC1 versus PC2 to see if you can see any "clusters" or "groups".

```
plot(svd1$v[,1],svd1$v[,2],col=2,ylab="2nd PC",xlab="1st PC")
```

One thing you can do is color them by different variables to see if clusters stand out.

```
plot(svd1$v[,1],svd1$v[,2],ylab="2nd PC",
     xlab="1st PC",col=as.numeric(pdata$study))
```

Another common plot is to make boxplots comparing the PC for different levels of known covariates (don't forget to show the actual data!).

```
boxplot(svd1$v[,1] ~ pdata$study,border=c(1,2))
points(svd1$v[,1] ~ jitter(as.numeric(pdata$study)),col=as.numeric(pdata$study))
```

## PCs versus SVs

What we have been plotting is not exactly the principal components.

```
pc1 = prcomp(edata)
plot(pc1$rotation[,1],svd1$v[,1])
```

To get the actual PCs you have to subtract the column means rather than the row means when normalizing.

```
edata_centered2 = t(t(edata) - colMeans(edata))
svd2 = svd(edata_centered2)
plot(pc1$rotation[,1],svd2$v[,1],col=2)
```

Despite this, it is most common for people to perform row-centering and then plot the singular vectors (sometimes labeling them PCs like I have done in this document)

## Outliers

What happens if we introduce a single outlying gene

```
edata_outlier = edata_centered
edata_outlier[1,] = edata_centered[1,] * 10000
svd3 = svd(edata_outlier)
par(mfrow=c(1,2))
plot(svd1$v[,1],col=1,main="Without outlier")
plot(svd3$v[,1],col=2,main="With outlier")
```

It turns out the new top singular vector is perfectly correlated with the outlying gene

```
plot(svd3$v[,1],edata_outlier[1,],col=4)
```

## Further resources

There are a large number of resources available about PCA and SVD but the lecture notes from [Advanced Statistics for the Life Sciences](#) are the best set of lecture notes focused on genomics currently available.

## Session information

Here is the session information

```
devtools::session_info()
```

It is also useful to compile the time the document was processed. This document was processed on: `r Sys.Date()` .

